

Final Project Phase I

Guidelines for Phase I Submission: For open ended questions, 1-3 bullet points should suffice for most answers. You do not need essay-length answers; however, there needs to be enough information, so that we can understand your topic and confirm that you have a cohesive and feasible topic. Make sure your answers are brief, but cohesive and answer all of the questions.

NOTE: Most of the points lost in this phase are due to not reading the instructions. Please make sure to read each question in its entirety.

Q1. Topic - 15 points

Please provide an overview of what your topic is going to be.

- 1. What topic have you chosen for your Final Project?**
 - a. How do government investments in transportation infrastructure influence traffic accidents and the usage of public transportation services between New York boroughs in 2022?
- 2. Why did you choose this specific topic and what are you looking to learn from the analysis?**
 - a. Pedro is very interested in traffic analysis and public transportation. Having studied abroad in Europe for a semester, it was eye-opening to him how different public and private transportation there differs from the US. As a result, this topic would demonstrate how our own government is using our tax dollars to create a transportation system in the most populous city in the country.
 - b. Owen is interested in public transportation and finding ways to reduce traffic. This project would allow him to demonstrate how to analyze traffic and public transportation using the skills he learned in 2316.
- 3. Explain some of the concrete insights you expect to gather from your data and/or hypothesis you expect to answer.**
 - a. Understanding the effectiveness in public transportation investments in ameliorating access and development of public transit.
 - b. Exploring the usage between private transportation methods and public public transportation methods and what the government can do to sway it one way or the other.
 - c. Investigating the best borough of New York to live in from a transportation standpoint.

Q2. Downloaded Dataset - 15 points

Please provide a brief overview of your downloaded dataset. This should demonstrate that you understand the data contained within the dataset.

1. **Provide the link (url) to your downloaded dataset.**
 - a. https://data.cityofnewyork.us/Public-Safety/Motor-Vehicle-Collisions-Crashes/h9gi-nx95/data_preview
2. **What are the dimensions of your downloaded dataset in terms of rows x columns and file size? Ex. 50,000 rows x 20 columns and 5.4mb. If your file is a .json file, state the file size (mb, gb, etc.).**
 - a. 1048576 rows x 30 columns and 19 mb.
3. **Briefly discuss the structure of your dataset. For .csv or table type datasets list out the column titles and give examples of the data contained within. For json data map out the dictionary and give examples of the data contained within.**
 - a. Columns: 'crash date', 'crash time', 'borough', 'zip code', 'latitude', 'longitude', 'location', 'on street name', 'cross street name', 'off street name', 'number of persons injured', 'number of persons killed', 'number of pedestrians injured', 'number of pedestrians killed', 'number of cyclist injured', 'number of cyclist killed', 'number of motorist injured', 'number of motorist killed', 'contributing factor vehicle 1', 'contributing factor vehicle 2', 'contributing factor vehicle 3', 'contributing factor vehicle 4', 'contributing factor vehicle 5', 'collision_id', 'vehicle type code 1 vehicle type code 2', 'vehicle type code 3', 'vehicle type code 4', 'vehicle type code 5'.
 - b. Crash date is formatted *MM/DD/YY*, Borough is a range of Manhattan, Brooklyn, Queens, Staten Island, and the Bronx. The columns with 'number' in the name have integer values to represent the number of people hurt / killed in the accidents.
4. **Please explain why you chose this specific dataset. How will this data be used in your analysis? Can insights be drawn from this data alone, or will it be combined with other data?**
 - a. This was chosen because it has an entirely comprehensive list of all the accidents that occurred in New York from 2016 to 2024. There are over 1 million data points and it would be excellent to work on my data manipulation skills. It also has some extra data on the type of accident it was and what cars were involved, which can be beneficial to see what type of cars are being driven on these government funded roads.

Q3. Web Requirement #1 (Web-scrape or HTML) - 15 points

Please provide a brief overview of your downloaded dataset. This should demonstrate that you understand the data contained within the dataset.

- 1. Provide the link (url) to your downloaded dataset.**
 - a. <https://www.budget.ny.gov/pubs/archive/fy22/ex/agencies/appropdata/TransportationDepartmentof.html#:~:text=Budget%20Highlights,from%20the%20FY%202021%20budget>.
- 2. Explain briefly how you plan to retrieve the data from this source.**
 - a. We will use BeautifulSoup and the requests module in order to extract the data from this source.
- 3. Briefly discuss the structure of your dataset. For .csv or table type datasets list out the column titles and give examples of the data contained within. For json data map out the dictionary and give examples of the data contained within.**
 - a. It's a HTML website where we will have to parse through the text data to get to the tables in the website to evaluate the information. An example of a table is: STATE OPERATIONS, where it shows the budget allocations to departments such as Bus Safety, Office of Passenger and Freight Transportation Program, and a General Fund
- 4. Please explain why you chose this specific dataset. How will this data be used in your analysis? Can insights be drawn from this data alone, or will it be combined with other data?**
 - a. This website had a very comprehensive overview of how the New York government uses the budget allocations to each department of transportation. From this, we can combine it with the other data to see how the spending on traffic safety per borough affects the number of traffic accidents.

Q4. Web Requirement #2 (API or JSON) - 15 points

Please provide a brief overview of your downloaded dataset. This should demonstrate that you understand the data contained within the dataset.

1. **Provide the link (url) to your downloaded dataset.**
 - a. https://data.cityofnewyork.us/Transportation/2022-Yellow-Taxi-Trip-Data/qp3b-zxtp/data_preview
2. **Explain briefly how you plan to retrieve the data from this source.**
 - a. We will use the requests module in order to extract the data from our API and make sure the data is converted to python from json formatting.
3. **Briefly discuss the structure of your dataset. For .csv or table type datasets list out the column titles and give examples of the data contained within. For json data map out the dictionary and give examples of the data contained within.**
 - a. This data highlights information on every taxi trip in 2022, with an example row: { 'vendorid': '1', 'tpep_pickup_datetime': '2022-01-01T00:33:43.000', 'tpep_dropoff_datetime': '2022-01-01T00:42:07.000', 'passenger_count': '1.0', 'trip_distance': '2.1', 'ratecodeid': '1.0', 'store_and_fwd_flag': 'N', 'pulocationid': '236', 'dolocationid': '42', 'payment_type': '1', 'fare_amount': '8.0', 'extra': '0.5', 'mta_tax': '0.5', 'tip_amount': '4.0', 'tolls_amount': '0.0', 'improvement_surcharge': '0.3', 'total_amount': '13.3', 'congestion_surcharge': '0.0', 'airport_fee': '0.0' }
4. **Please explain why you chose this specific dataset. How will this data be used in your analysis? Can insights be drawn from this data alone, or will it be combined with other data?**
 - a. This data is useful because with taxi data, we can see how the numbers in each borough are affected in relation to the government spending on public transportation and safe routes for taxis and public transportation vehicles. We can see if more people are using taxi services due to higher government investment.

Q5. Additional Datasets - 10 points

If you have found any datasets beyond the three required, please describe them below: (If you do not plan to use any additional datasets please simply write N/A)

1. **Provide the links for any additional datasets you might use.**
 - a. <https://ibo.nyc.ny.us/fiscalhistory.html>
2. **Explain how you will retrieve data from these sources, and how this data is going to be used for your analysis**
 - a. In this website, there is an Excel file under Agency Expenditures that gives detail on the spending per borough and each department. This is beneficial because we could possibly compare the borough spending and see how one borough might be more efficient in their spending.

Q6. Inconsistencies - 15 points

Please list at least 3 inconsistencies you have found in your datasets, and how you plan to address each of them.

1. In the NYC Motor Collisions CSV, there is various missing data in the Contributing Factor columns. These would be filled with 'N/A'.
2. In the NYC Fiscal History CSV, there is missing data in the amounts allocated to certain departments. These would be filled with 0 to represent a non-allocation of funds that year to that borough.
3. In the Taxis API, the format of the dates is a str of the entire date including the year. This would be formatted to be an int of the day into the year (ex: January 3rd is int 3 and February 1st is int 32)

Q7. About Your Analysis - 10 points

Provide a BRIEF list of steps of how you plan on performing your analysis and the way you will gather/present your findings. (Non-technical, high-level overview)

1. We are going to compare how the amount of government spending in the different boroughs of NYC impacts the amount of accidents and public transportation issues
2. We will sort through and clean our data from the first source so that we use info regarding the boroughs and the amount of accidents that occurred there.
3. We will sort through and clean our data from the second source so that we use info regarding the spending allocated to each borough for improvements to the public transportation systems.
4. We will sort through and clean our data from the third source so that we can use info from our taxis API.
5. We will sort through and clean our data from the fourth source so that we use info regarding the boroughs and the spending for each.
6. We will then use the cleaned data from our sources to come up with a hypothesis and a correlation that explores the comparison between government spending in the boroughs and the amount of accidents in each.
7. We will compile the data and our arguments/conclusion and present it

Q8. About You - 5 points

List the names of each of the members of the group working on this project. If you are working alone, there should be one name listed.

Team Member 1: Pedro Kawamura

Team Member 2: Owen Seibold

Each member of the group should initial below to indicate that you acknowledge this statement:

I affirm that all of the work in this project will be done by me/my team and is not duplicated from any other source. In addition, any references that I use or code that I choose to model after will be appropriately credited and referenced in my project.

Team Member 1 Initials: PK

Team Member 2 Initials: OS

Total - 100 points